MiCucumber: in silico design and evaluation of mitochondrial primers for sea cucumber (Holothuroidea) eDNA metabarcoding

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Abstract

Environmental DNA metabarcoding of sea cucumbers requires markers that combine broad primer compatibility with informative internal sequence variation. We evaluated five mitochondrial 16S windows on a curated panel of 174 species and independently screened mitochondrial 12S and COI. Each retained ordinary/D alternative shares reference endpoints, with at most eight concrete variants per D oligonucleotide. On the common 151-species 16S panel, SD covered 115 species, whereas LD gave the most exact paired matches (23 species). Performance varied by lineage: LD covered 71/77 evaluable Holothuriida but 0/22 Elasipodida and 0/5 Apodida on its own panel. Primer-free inserts contained six distinct interspecific identity-conflict pairs. Genus-blocked sequence selection quantified compatibility conditional on the predefined 16S windows. Among retained 12S D assays, XLD covered 33/53 species with seven exact matches. The refined 145 bp COI XS and XSD covered 20/57 and 43/57 species, with two and three exact matches and no strict background hits among 2,216 reference records. COI LD covered 103/283 species. Mismatch-gradient analysis of all 29 pairs showed that 16S SD matched 0, 24 and 130/174 species at zero, one and two mismatches per primer. Fifteen taxonomically ordered internal-barcode matrices comprised 130,599 within-window species comparisons. The resulting sequence collection and reproducible analyses support marker choice by length, lineage and binding properties. Released-assay estimates use the design panels, while the genus-blocked diagnostic omits the formal thermal and background gates.

Keywords: Holothuroidea; environmental DNA; mitochondrial 16S; primer degeneracy; barcode discrimination; grouped holdout

Introduction

Sea-cucumber molecular identification has an established basis in mitochondrial reference libraries. Uthicke et al. (2010) evaluated COI barcodes for commercially harvested species, while Byrne et al. (2010) used COI and 16S to clarify relationships and taxonomic assignments within Stichopodidae. For processed products, Wen et al. (2010, 2011) developed PCR–RFLP and forensically informative nucleotide sequencing (FINS) approaches based on mitochondrial 16S. These applications support retaining explicit species identities and source accessions when designing broadly compatible primers.

Xing et al. (2021) developed a 257 bp COI mini-barcode and combined it with next-generation sequencing to authenticate processed sea-cucumber products. Their study provides a sea-cucumber precedent for shortening a barcode and recovering multiple ingredients by sequencing. Environmental samples add a different template mixture and sampling context, so MiCucumber evaluates target compatibility, background matching and reference-insert discrimination on its own frozen panels.

Environmental DNA metabarcoding uses short markers to characterize biological assemblages from mixed DNA. Marker selection combines conserved primer-binding regions with informative variation between them. MiFish exemplifies this approach for fish 12S (Miya et al., 2015), while ecoPrimers

formalizes the joint evaluation of taxonomic coverage and discrimination (Riaz et al., 2011). These principles motivate Holothuroidea 16S candidate comparison without transferring performance estimates from other taxa or loci. Anthozoan 28S work additionally illustrates how lineage-specific binding-site variation can motivate controlled primer degeneracy (McCartin et al., 2024).

A primer matching many database records may primarily reflect redundant sampling. A region with a high percentage of matched species can also be supported by a smaller or taxonomically different panel than another region. We therefore ask how controlled ordinary/D alternatives differ in coverage and exact matching on the same species set, how much discrimination their internal markers provide, and whether sequence selection generalizes to genera omitted from its training fold.

Materials and methods

Data acquisition and reference curation

The frozen source project contains 1,848 target records and 5,600 non-target records. Taxonomic and quality curation retained one representative for each of 174 species, using sequence completeness and source classifications rather than primer-match success. The source repository provides accession-level audit and classification files. Alignments use MAFFT 7.526 (Katoh and Standley, 2013).

Alignment and reference coordinates

Reference positions are 1-based and inclusive on ungapped mitochondrial 16S sequence AB525437.1; they refer to the extracted gene, not whole-mitogenome coordinates.

Binding-window evaluation requires that both windows span the reference interval, contain at least 90% of the corresponding reference-site count as concrete A/C/G/T, and contain no ambiguous nucleotide. Alignment gaps are allowed. Missing or uncertain binding windows are excluded from that region's denominator rather than scored as primer failures. The intersection of available accessions defines the common panel of 151 species. This is an available-data intersection, not a random sample of global diversity.

The numerical results are reported in Table 1.

The reference accession AB525437.1 belongs to Apostichopus japonicus. Coordinates for 16S, 12S and COI are one-based positions within the separately extracted gene sequences.

Candidate generation and chemical screening

Within S–XL, ordinary and D sequences are selected independently at shared reference start and end positions. The D sequence can differ from the ordinary sequence at concrete as well as ambiguous positions. Released D oligonucleotides have no more than eight concrete variants per primer and confine ambiguity to the 3' half. The source design excludes homotriplet ends and homopolymers longer than four nucleotides and requires a G or C within the final three positions. Primer3 (Untergasser et al., 2012) evaluations use 50 mM monovalent salt, 1.5 mM Mg2+ and 0.6 mM dNTP, under two separate concentration scenarios: 50 nM per concrete variant, and 200 nM total per primer direction divided equally among its concrete variants. The S–XL gates require a maximum predicted hairpin/dimer structure Tm below 55 °C and a forward/reverse duplex-Tm range midpoint difference no greater than 5 °C. The 16S sequences were selected on the curated design panel through target, terminal, chemical and background screening. Their own-panel and common-panel scores therefore describe performance on references that participated in design.

Matching models and common panels

Coverage requires no more than two mismatches per primer and IUPAC compatibility in each terminal three nucleotides. Exact matching requires zero mismatches at both binding sites. S–XL paired target scans allow total products of 150–500 bp with at least 50 bp between primers. XS scans allow 70–500 bp products with at least 20 bp between primers. Both sequence orientations are considered. Species contribute equally; per-order counts accompany aggregate values.

Released background screens allow an internal spacing of 0–1,000 bp. Strict screens retain the terminal constraint for both models. Permissive screens allow up to three mismatches per primer with Watson–Crick pairing alone and up to two when guanine–thymine pairing is also allowed. These counts describe reference records, not experimental amplification frequencies. C++ and Python implementations were previously compared on the same frozen data to check target and strict-background hit sets. Such implementation agreement is distinct from held-out reference evaluation.

Primer-free barcode discrimination

For each available representative, we extract homologous alignment columns strictly between the forward-primer end and reverse-primer start, remove gaps, and exclude ambiguous internal inserts. These reference inserts exclude both primers and are not assumed to be observed PCR products. Inserts are grouped by exact sequence identity independently of the formal primer's matching result. Identity uniqueness is assessed against all available unambiguous inserts for that window, including species outside the common panel. The covered-and-unique count requires both a paired primer match and an identity-unique insert, with the full comparison-panel denominator retained.

Unit-cost Levenshtein distances are calculated between all pairs of unambiguous inserts. For each species we report its nearest other-species distance and all tied accessions. Distance implementation was cross-checked against an independent dynamic-programming algorithm in the initial workflow. One representative per species cannot estimate within-species diversity or a barcode gap. Identity uniqueness therefore describes the frozen reference collection rather than empirical taxonomic-assignment accuracy. Apparent identity conflicts are preserved for source review. A sensitivity table removes all members of each same-window identity conflict without changing the primary panel.

Genus-blocked sequence-selection test

For each window, genera are sorted and linked when they share an identical complete representative sequence. Blocks are shuffled using Python's random generator with seed 20261030 and assigned cyclically to five folds. Consequently, no genus or identical complete representative crosses training and test sets. Fold sizes can differ substantially where genera are unevenly sampled.

Within each training fold, observed local sequences within five alignment columns of the frozen binding intervals supply candidates of fixed oligonucleotide lengths. Up to 24 frequent admissible concrete seeds per side enter deterministic greedy growth with at most three ambiguity-expansion steps. Ambiguity is confined to the 3' half, degeneracy is limited to eight variants per oligonucleotide, a concrete G/C is required within the final three positions, and literal five-nucleotide homopolymers are excluded. Training paired coverage is maximized, followed by exact matching and lower summed degeneracy, with lexical tie-breaking. Released primer sequences and global candidate pools do not seed this training step.

Held-out species are scored by full raw-sequence paired scans with the same target size, mismatch and terminal rules. Each available species is tested once per window. Fold candidate construction failure would yield zero detections for that fold and empty sequence fields. This diagnostic selector does not repeat the original thermal or background selection gates. Region discovery and source-panel

curation preceded splitting; accordingly, the test is conditional on predetermined regions and does not externally validate complete window discovery. Its fold sequences are not replacement recommendations.

Independent 12S and COI candidate generation

The search of NCBI Nucleotide retrieved 6,168 distinct accessions across both loci. Annotated features were extracted in strand orientation and screened for length, ambiguity and nuclear or pseudogene annotations. Independent design panels contained 90 named species for 12S and 357 for COI. Candidate binding windows were evaluated on each gene's own alignment. Ordinary primers were chosen from the concrete variants of each retained D pair at the same reference endpoints. Each oligonucleotide has degeneracy ≤8. Within each size class, 12S prioritized exact matches among candidates covering at least 80% of the best eligible coverage count. COI maximized coverage among candidates retaining at least 95% of the largest evaluable panel. Raw paired scans allowed 100–500 bp target products with an internal insert of at least 50 bp. The frozen backgrounds contain 438 reference records for 12S and 2,216 for COI. Thresholds were ≤10 strict hits per model and ≤50 relaxed hits per model. The COI XS window was subsequently refined by shifting binding positions and jointly selecting ordinary and D sequences at fixed shared endpoints. The adopted window spans gene-relative positions 779–923, with F 779–798 and R 907–923. Independent seed-and-extend matching verified 57,550 record/model comparisons across the 20 released 12S and COI primer pairs.

Independent implementation checks

The frozen computational bundle contains input hashes, source sequences, window panels, scripts, per-species predictions, fold candidates and aggregate tables. Numerical analysis uses Python's standard library; plotting uses matplotlib. Validation checks denominators, fold sizes, genus separation, identical-sequence grouping, input hashes and summary totals. Source audits remain available in the repository. The present manuscript was generated from the committed result tables, with input hashes recorded in its provenance file.

Mismatch sensitivity, product lengths and taxonomic sequence identity

Following the diagnostic approach of McCartin et al. (2024, pp. 6 and 11–13; Fig. 3), all 29 released primer pairs were rescanned on both strands of their frozen target and background sequences. Mismatch thresholds of zero, one and two apply separately to each primer, with IUPAC compatibility treated as a match under Watson–Crick pairing. Each threshold was evaluated with and without requiring compatibility at the final three 3' bases of both primers. Ambiguous binding sites were excluded. Target product-length limits reproduce the released rules: 70–500 bp with an insert ≥20 bp for 16S XS, 150–500 bp with an insert ≥50 bp for the remaining 16S assays, and 100–500 bp with an insert ≥50 bp for 12S/COI. Background inserts were allowed from 0 to 1,000 bp. Target denominators contain one representative per evaluable species per window; background denominators are 5,600, 438 and 2,216 accession records for 16S, 12S and COI. Counts therefore describe different units and are not amplification proportions in a mixed sample. The released guanine–thymine and more permissive background models remain separately reported.

For sequence-identity matrices, primer-free inserts were extracted between the aligned reference endpoints from each window's evaluable panel. Inserts containing ambiguous nucleotides were excluded; the existing curated gene alignments were retained. Identity equals the number of equal A/C/G/T columns divided by the number of columns containing at least one base. A base opposite a gap is a difference, and double-gap columns are ignored. Matrices were sorted by the source order assignment, genus and species. Ordinary/D alternatives share the same internal region, so one matrix represents each length. Exact ungapped insert identity was also counted independently. Between-

species similarities were stratified into same genus, same order/different genus, and different orders. These available-panel summaries do not provide within-species variation, a barcoding-gap test, or matched cross-gene performance estimates. Product distributions count each unique accession/product-span combination once at each threshold; target products are restricted to the evaluable window panel.

The new C++ exhaustive scan was compared with the released exact and strict Watson–Crick summaries in 87 checks. An independent Python regex-seed matcher verified 1,578 threshold/terminal flags on deterministic hit and non-hit subsets spanning every assay and both target/background roles. A further 465 sampled pairwise identities were recalculated independently.

Results

Released assays on their available panels

On their available 16S panels, SD covers 130/174 species, MD covers 89/174 with 24 exact matches, and LD covers 82/174 with 29 exact matches (Table 2).

Own-panel proportions cannot by themselves establish a ranking across lengths because both denominator size and lineage composition differ. Strict and permissive background metrics remain distinct in the repository's released comparison table.

Equal-denominator comparison

The common panel includes 151 species, compared with 166 evaluable XS species, 174 in S–L and 152 in XL. On the common panel, SD covers 115 species but has no exact paired matches; MD covers 79 with 22 exact, and LD covers 73 with 23 exact. The short XS covers 69 with no exact paired matches. These contrasts separate mismatch-tolerant breadth from exact template compatibility. In the own-window panel, LD covers 71/77 Holothuriida, 8/36 Dendrochirotida and 0/22 Elasipodida, demonstrating strong lineage dependence.

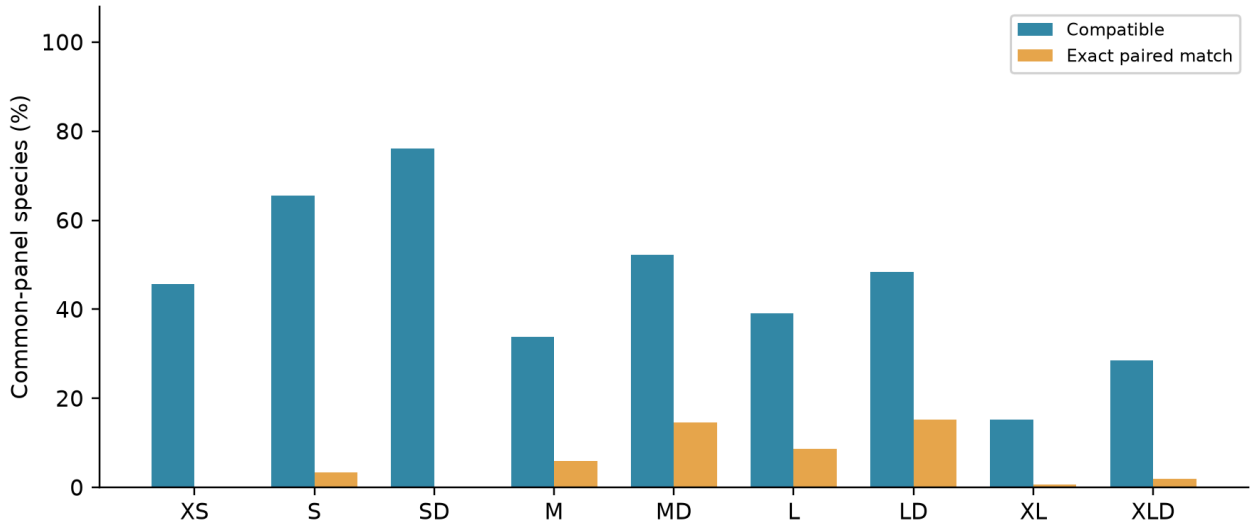


Figure 1. Compatibility and exact paired matching on the common species panel. All bars use the same denominator.

Reference-insert discrimination

Identity-unique reference inserts range from 154/166 species in XS to 146/152 in XL; the common-panel covered-and-unique counts further require compatible primers (Tables 3–4).

The conflict inventory contains six distinct interspecific pairs, corresponding to 18 same-window identity groups across the five regions. Three pairs recur in every window: Astichopus multifidus–Isostichopus macroparentheses, Scotoplanes clarki–Protelpidia murrayi and Stichopus ocellatus–Stichopus hermanni. These are reference-set observations, not a determination of their biological cause. Taxonomic synonyms, source identification, sequence annotation and genuine marker sharing require distinct follow-up assessments. All accessions and sequences are retained in the supplemental conflict table. Reporting conflicts avoids counting every compatible template as a uniquely identifiable species. The conflict-exclusion sensitivity analysis is descriptive and does not assert that any source record is erroneous.

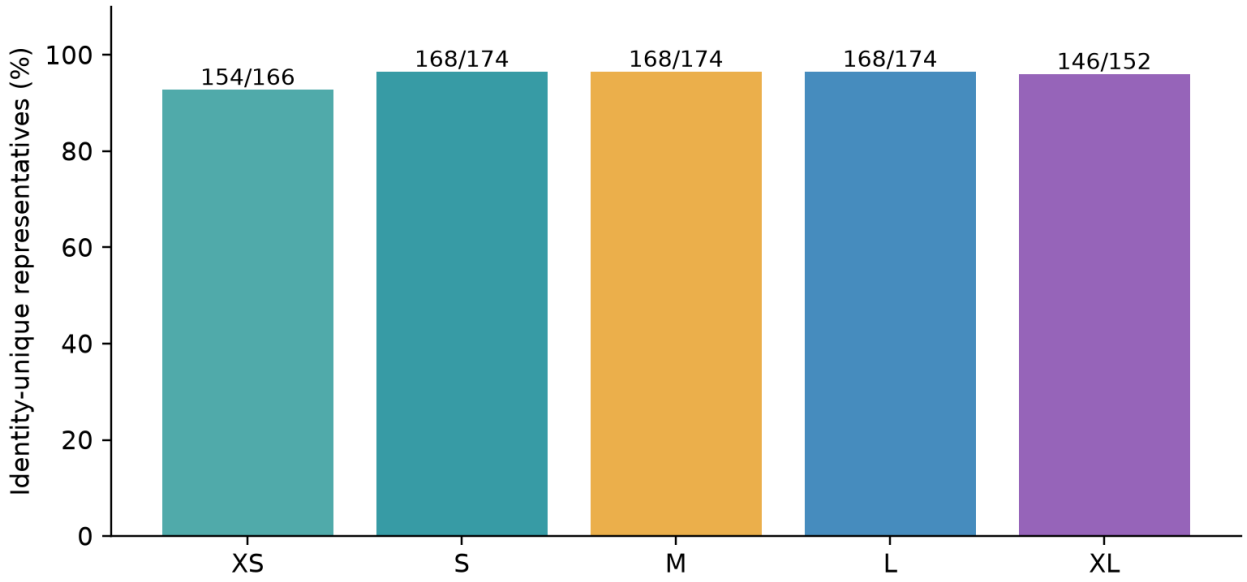


Figure 2. Identity-unique species among all available unambiguous reference inserts in each window. Labels show numerator and denominator.

Conditional genus holdout

Across the five held-out folds, diagnostic sequence selection covers 156/166 XS species and 170/174 S species, with 64 and 39 exact matches, respectively (Table 5).

Each row pools five folds. Fold membership, genus counts, candidate sequences and training/test sizes are available in the computational bundle. A high compatible count can coexist with low exact matching. Because chemistry and background constraints were not applied to these diagnostic candidates, a larger test percentage is not evidence of a superior released assay. Conversely, a poor score can reflect the constrained heuristic rather than the absence of any useful primer in a region.

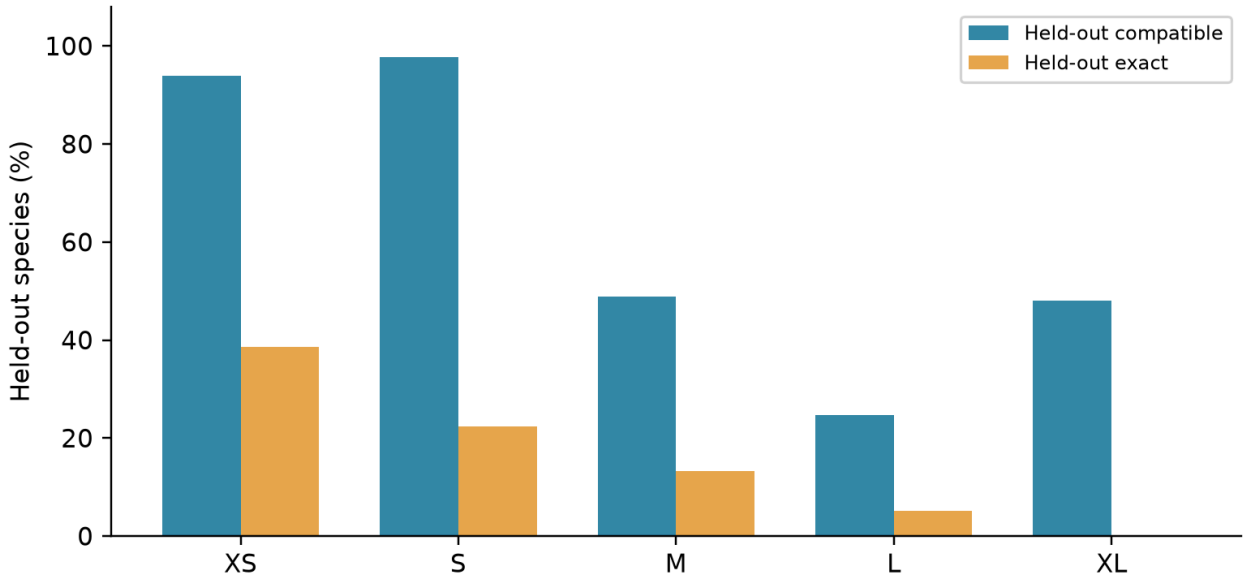


Figure 3. Conditional genus-blocked test predictions. Fold-specific training sequences differ from the released ordinary/D assays.

Independent 12S and COI screens

Among retained 12S D pairs, XLD has the largest covered-species count and coverage proportion on its available panel: 33/53 species, with seven exact paired matches (Table 6). MD covers 28/90 species, with four exact matches. Because the windows have different evaluable species, these values describe their available panels.

The adopted COI XS and XSD share a 145 bp reference window and 108 bp primer-free reference insert. XS covers 20/57 species with two exact matches; XSD covers 43/57 with three. Both have zero strict hits in either background model. On this background, their permissive Watson–Crick/guanine–thymine counts are 12/1 and 32/1. Relative to the preceding XS sequence, the adopted ordinary pair loses five previously covered species and gains fourteen; XSD loses five and gains seven. These changes in species membership are listed in the current README.

The COI L window has 283 evaluable species compared with 55–57 for XS through M. Its ordinary and D pairs cover 30 and 103 species, with zero and eleven exact matches. Reference-window availability and compatibility therefore both influence the covered-species count. The common-panel discrimination and genus-blocked sequence-selection analyses apply to 16S.

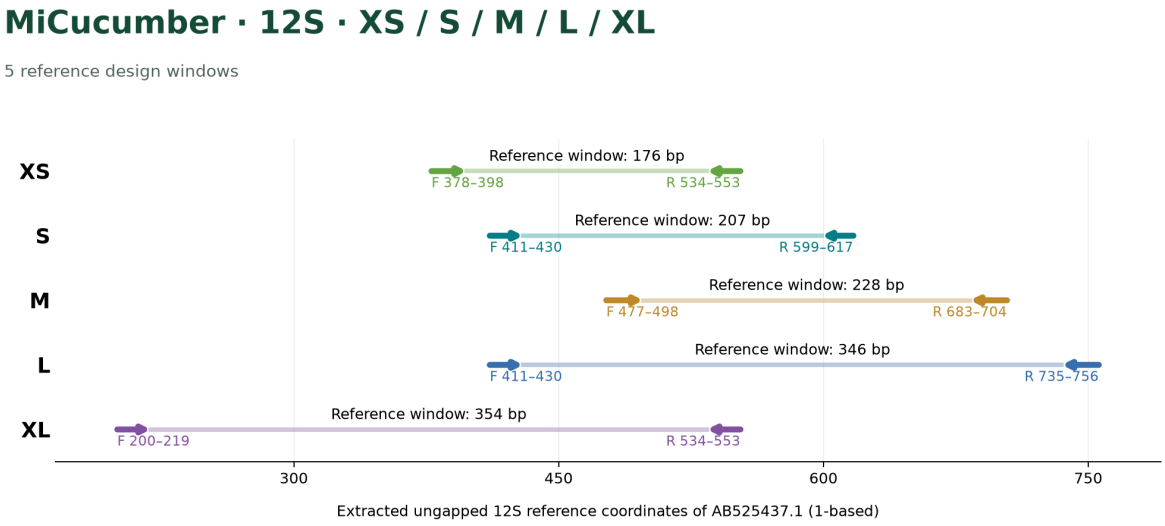


Figure 4a. Ordinary and degenerate primer windows on the 12S reference.

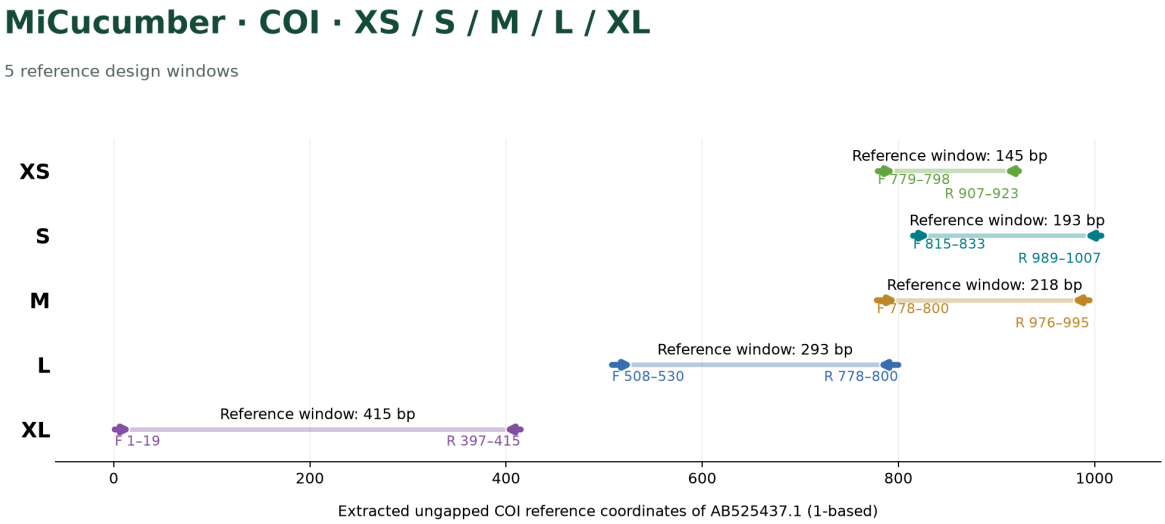


Figure 4b. Ordinary and degenerate primer windows on the COI reference.

Anth-28S-inspired diagnostic results

The mismatch series separates exact compatibility from tolerance-dependent coverage. On their own available panels, 16S SD matched 0, 24 and 130 of 174 species at zero, one and two mismatches per primer; requiring terminal-three compatibility at the two-mismatch threshold retained all 130. In contrast, LD matched 29, 60 and 84 species, decreasing to 82 after the terminal constraint. For 12S XLD, the series was 7, 16 and 33 of 53 species, with 33 retained by the terminal rule. COI XSD matched 3, 18 and 44 of 57 species, decreasing to 43; COI LD decreased from 122 to 103 of 283 species when terminal compatibility was required at the two-mismatch threshold. These differences show why relaxed mismatch totals cannot replace the released strict coverage measure.

At up to two Watson–Crick mismatches without the terminal constraint, the frozen background scan predicted one record for each of 16S XS and S and none for the other 27 pairs. This restricted model does not replace the reported guanine–thymine or higher-mismatch screens. The product-length plot

presents these finite-panel predictions without assuming size selection will separate target and background products.

The 15 identity matrices contain 130,599 distinct within-window species comparisons. Exact insert uniqueness was 154/166 species for 16S XS and 168/174 for S, M and L; the XL panel contained 146/152 unique representatives. For 12S XL, 53/53 available inserts were unique; COI XL contained 62/62 unique inserts. The COI L identity analysis retained 278 of 283 evaluable species after excluding five ambiguous internal inserts, of which 272 had unique inserts. Complete sequences can therefore differ in primer evaluability and usable internal-barcode status. Neither smaller-panel uniqueness nor higher average divergence alone establishes a better assay.

Table 8 reports the mismatch-gradient counts for all 29 assays; Table 9 summarizes the internal-barcode panels.

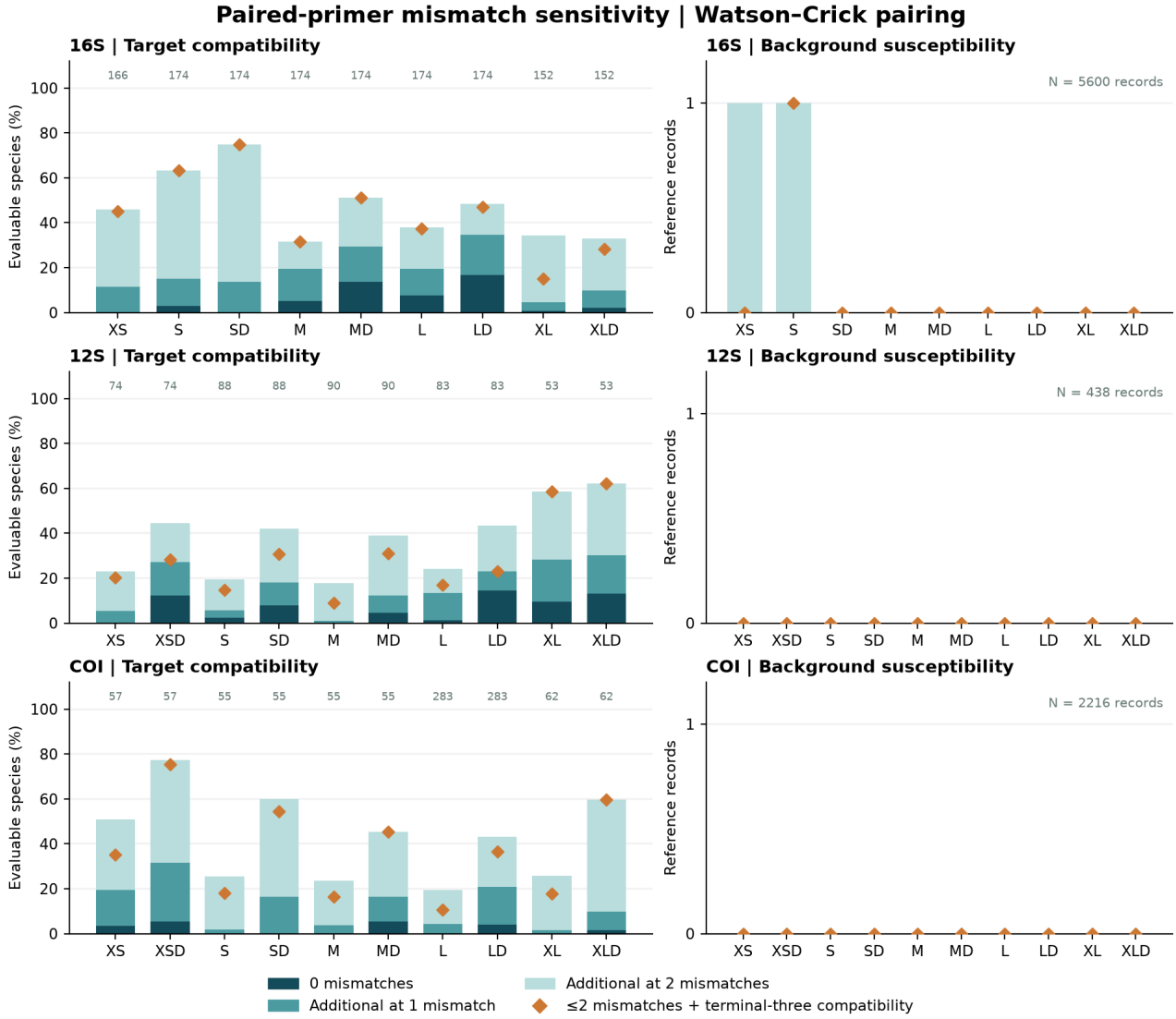


Figure 5. Mismatch sensitivity for the 29 released assays. Stacked bars show cumulative zero-, one- and two-mismatch matches, with thresholds applied to each primer separately. Orange symbols require terminal-three compatibility at the two-mismatch threshold. Target annotations give evaluable species denominators; backgrounds count accession records. Only Watson–Crick pairing is shown.

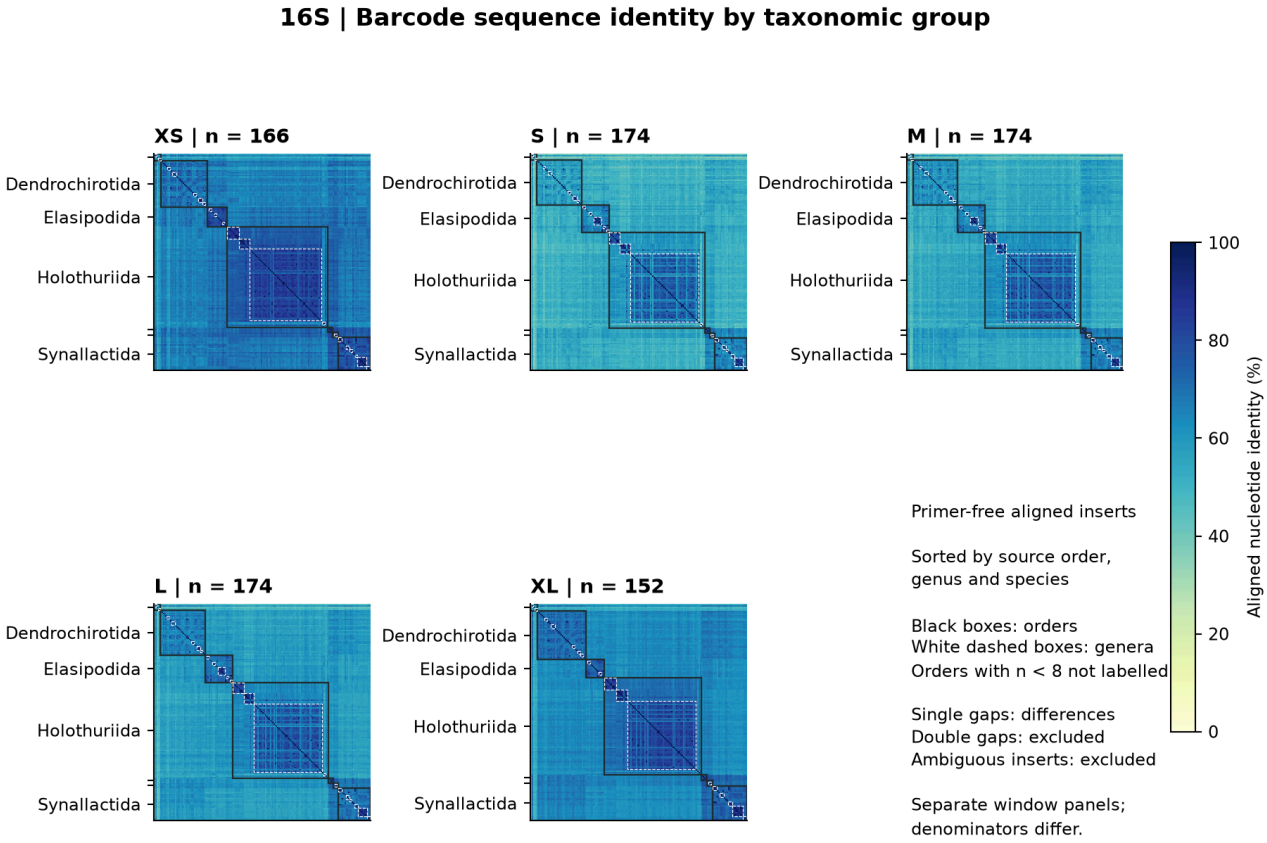


Figure 6. 16S primer-free aligned sequence identities, sorted by source order, genus and species. Black boxes delimit orders and white dashed boxes delimit genera. All available unambiguous inserts are included regardless of released-primer compatibility. Single gaps count as differences and double gaps are excluded.

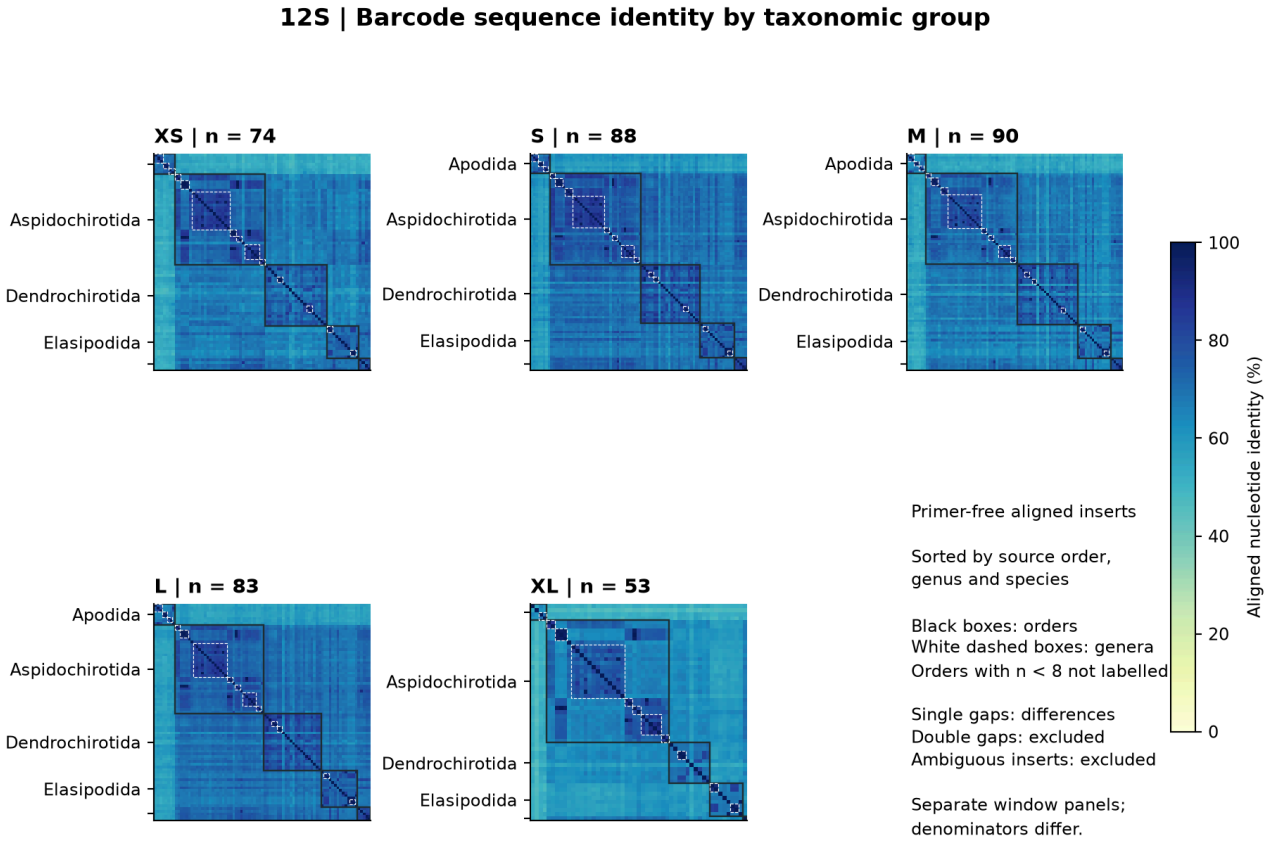


Figure 7. 12S primer-free aligned sequence identities, using the same definition as Figure 6. Each window uses its own available panel and species order.

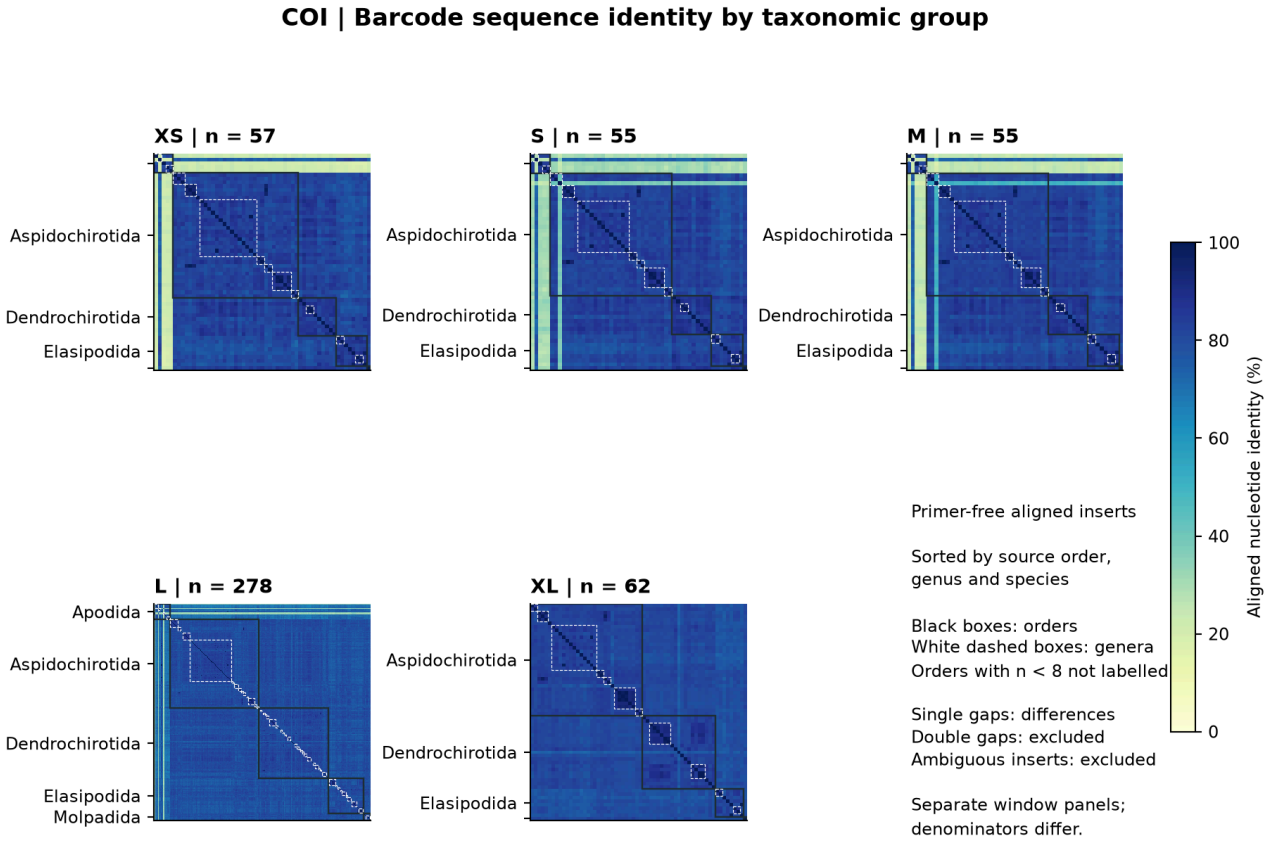


Figure 8. COI primer-free aligned sequence identities, using the same definition as Figure 6. Five ambiguous internal inserts were excluded from the L window; its identity denominator is 278 rather than the binding-evaluable 283.

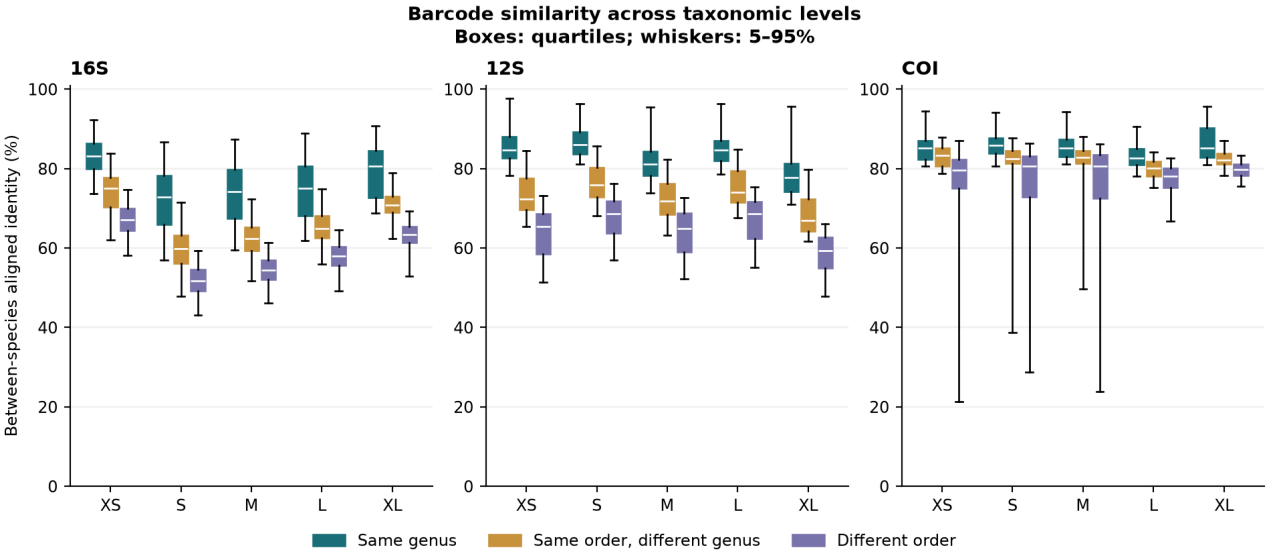


Figure 9. Between-species aligned identity by taxonomic relationship. Boxes span quartiles; whiskers show the fifth and ninety-fifth percentiles. Pairwise observations share species and are descriptive, not independent statistical replicates. One representative per species precludes assessment of a within-/between-species barcoding gap.

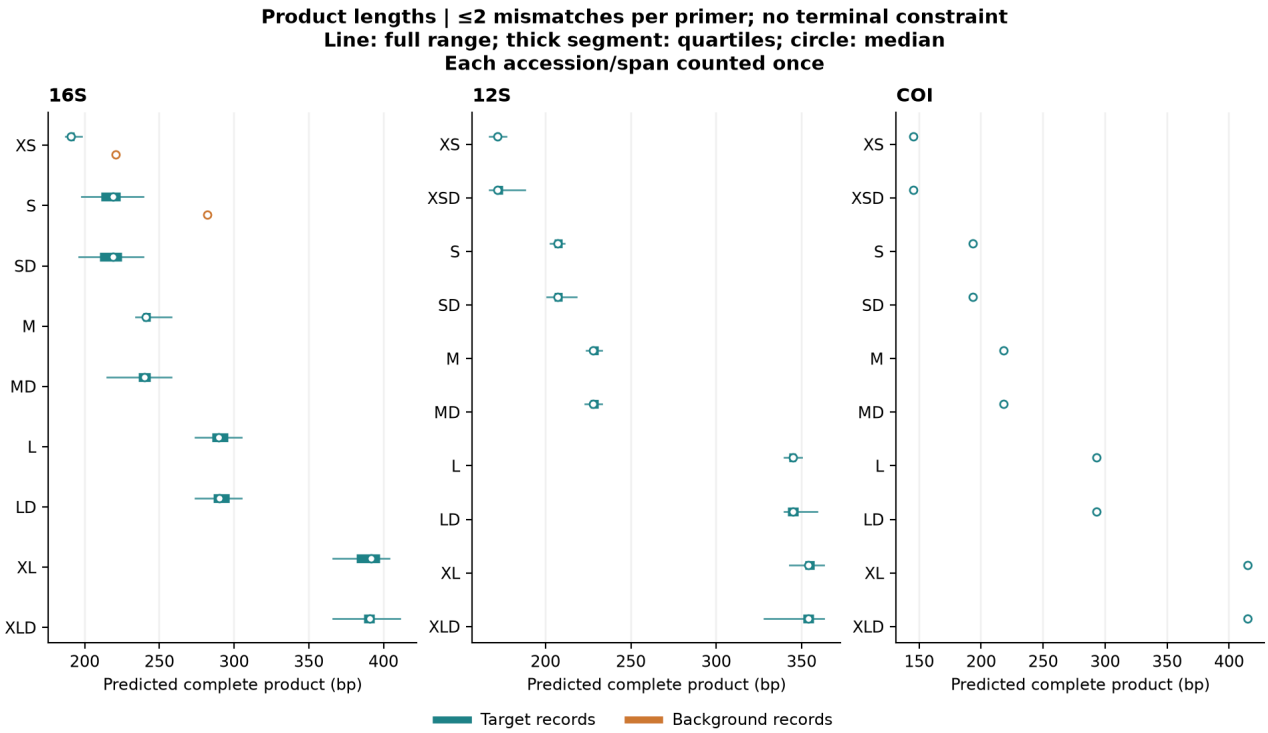


Figure 10. Predicted complete product lengths at up to two mismatches per primer, without a terminal constraint. Lines show full ranges, thick segments show quartiles and circles show medians of unique accession/span combinations. Background matches occur only for 16S XS and S in this restricted model. Product-length separation is not assumed.

Discussion

The mismatch-gradient analysis further shows that high two-mismatch compatibility can coexist with very few exact or one-mismatch matches. For example, 16S SD has no exact matches but 130/174 two-mismatch matches on its available panel. These are complementary diagnostics, and should not be collapsed into a single assay ranking. Taxonomically sorted identity matrices expose shared and divergent internal sequences, while the separate ambiguity filter keeps primer-site evaluability distinct from barcode usability.

The analysis separates three questions: whether a primer pair matches the target reference, whether the paired sites match exactly, and whether the internal sequence distinguishes that reference from other named species. On the common 151-species panel, the largest predicted coverage was 115/151 (SD), and the largest number of exact paired matches was 23/151 (LD). These properties need not favor the same design. Coverage improvements from controlled degeneracy should therefore be interpreted alongside exact matches, lineage changes and non-target susceptibility rather than a single aggregate percentage.

Reference availability is a substantive part of the result. Common-panel comparison removes changing denominators but limits the taxa represented. Own-panel analysis increases available taxonomic information while making raw percentages less directly comparable across regions. Order-level files expose which lineages contribute to each result, and nearest-neighbor distances show whether sequence uniqueness depends on very small differences. Identity conflicts should be reviewed against source records before being interpreted as biological limits of the marker.

The genus-blocked diagnostic prevents direct reuse of a genus across training and testing and binds identical full representatives together. It is nevertheless narrower than independent validation of the complete discovery pipeline: the windows and curated species collection were chosen before splitting.

A fully nested region-discovery design or external accession set unused in any design stage would support broader generalization claims. Fold candidates also lack the original chemical/background selection and allow local alignment offsets; they cannot replace the fixed-coordinate formal assays on the basis of these tests.

One-reference-per-species scoring controls database redundancy but does not cover all haplotypes, populations or sequencing errors. Exact reference-insert uniqueness is not classifier accuracy, and compatibility models do not estimate competitive amplification in mixed DNA. Background libraries are finite and strongly shaped by available source data. Direct benchmarking against verified published taxon-compatible 16S primers, broader background sampling and repeated grouped splits remain additional computational analyses rather than completed results. These boundaries define the interpretation of this computational resource.

Published barcoding results help interpret the reference identity conflicts. Uthicke et al. (2010) reported conspecific COI variation of 1.3% and congeneric divergence of 16.9%, with a small overlap and some morphologically disparate specimens sharing a barcode. Their observations motivate source and taxonomic review of shared sequences, including the 16S conflicts documented here. Byrne et al. (2010) further showed how combined COI and 16S evidence can clarify specimen assignments in Stichopodidae. A fixed short-window collision therefore warrants evaluation together with specimen identity and independent sequence evidence.

Low degeneracy also has an echinoderm-specific methodological precedent. Hoareau and Boissin (2010) designed hybrid COI primers using 310 species, reporting fourfold degeneracy compared with at least 48-fold for their standard degenerate designs, and amplification in all 123 tested species. Their assays targeted more than 500 bp. MiCucumber applies a low-degeneracy constraint to shorter candidate windows and evaluates every retained pair against its own target and background references. The primer architecture, amplicon length and tested material determine the scope of each performance estimate.

Priorities for assay evaluation

Intended use	Computational priority	Evidence
Broad 16S compatibility	SD	115/151 common-panel species covered
Exact 16S matching	LD	23/151 common-panel species matched exactly
Additional 12S compatibility	XLD	33/53 covered and 7 exact on its available panel
Short COI window	XS / XSD	145 bp; 20/57 and 43/57 covered
Longer COI coverage	LD	103/283 covered on its available panel

The broad-compatibility priority for 16S SD uses the mismatch-tolerant model; its common-panel exact-match count is 0/151. Priorities within 12S and COI use each window’s available panel, so they do not establish an equal-denominator ranking across lengths.

Conclusions

MiCucumber provides five controlled mitochondrial 16S windows for computational comparison in sea cucumbers. Common-panel results, internal-barcode identity and conditional genus holdout reveal complementary aspects of candidate performance. The 145 bp COI XS/XSD window adds a short-marker choice with 20/57 and 43/57 compatible species and no strict background hits on the frozen panel. For 16S, limited Apodida and Elasipodida compatibility remains a substantial lineage constraint. The released sequences and source-linked tables support length and degeneracy choices within the available reference coverage.

Data availability and reproducibility

All primer sequences, frozen reference inputs, analyses and this preprint are available at https://github.com/GUIBA-EX/MiCucumber. The common-panel evaluation bundle is under results/computational_evaluation and the mismatch/identity diagnostic bundle is under results/anth28s_evaluation; this manuscript's Markdown, PDF, build script and provenance are under publications. No new database retrieval is needed to reproduce the numerical analyses from the frozen inputs.

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Versioned preprint and associated sequence files: https://github.com/GUIBA-EX/MiCucumber/releases/tag/preprint-v1.5

Table 1. Reference binding coordinates and evaluable species.

Spans include both primer-binding regions; actual products vary among species.

Window	F positions	R positions	Span bp	Available species
XS	1042–1059	1213–1232	191	166
S	954–977	1156–1179	226	174
M	921–942	1148–1165	245	174
L	868–891	1152–1169	302	174
XL	781–800	1147–1176	396	152

Table 2. Released 16S own-panel target results and strict background-record counts.

[1] Counts give Watson–Crick pairing alone first, followed by the model that also allows guanine–thymine pairing.

Assay	Own-panel N	Covered	Exact	Strict background hits[1]
XS	166	75	0	0 / 0
S	174	110	5	1 / 3
SD	174	130	0	0 / 5
M	174	55	9	0 / 2
MD	174	89	24	0 / 15
L	174	65	13	0 / 0
LD	174	82	29	0 / 0
XL	152	23	1	0 / 0
XLD	152	43	3	0 / 10

Table 3. Released assay outcomes on 151 common species.

Covered + unique means compatible paired primers and an identity-unique reference insert against the entire available same-window barcode set.

Assay	Covered	Exact	Covered + unique
XS	69/151	0/151	63/151
S	99/151	5/151	95/151
SD	115/151	0/151	109/151
M	51/151	9/151	47/151
MD	79/151	22/151	73/151
L	59/151	13/151	59/151
LD	73/151	23/151	73/151
XL	23/151	1/151	23/151
XLD	43/151	3/151	43/151

Table 4. Unambiguous internal reference inserts.

Nearest ≤1 is the number of representatives with a different species at edit distance zero or one. N can be smaller than the binding-window panel when an internal insert is ambiguous.

Window	N	Unique	Insert bp	Median nearest edit	Nearest ≤1
XS	166	154	148–165	10.0	20
S	174	168	148–191	20.0	14
M	174	168	175–218	22.0	14
L	174	168	230–274	26.5	14
XL	152	146	313–361	31.5	10

Table 5. Out-of-fold compatibility from training-selected diagnostic pairs.

These are not the released assay scores in Tables 2–3.

Window	Test N	Covered	Exact
XS	166	156 (93.98%)	64 (38.55%)
S	174	170 (97.70%)	39 (22.41%)
M	174	85 (48.85%)	23 (13.22%)
L	174	43 (24.71%)	9 (5.17%)
XL	152	73 (48.03%)	0 (0.00%)

Table 6. Predicted compatibility and strict background-record counts ([1] Watson–Crick pairing alone / also allowing guanine–thymine pairing) in the independent 12S and COI panels.

Gene	Assay	Span bp	Covered/N	Exact/N	Strict background hits[1]
12S	XS	176	15/74	0/74	0/0
12S	XSD	176	21/74	9/74	0/0
12S	S	207	13/88	2/88	0/0
12S	SD	207	27/88	7/88	0/0
12S	M	228	8/90	0/90	0/0
12S	MD	228	28/90	4/90	0/5
12S	L	346	14/83	1/83	0/0
12S	LD	346	19/83	12/83	0/0
12S	XL	354	31/53	5/53	0/1
12S	XLD	354	33/53	7/53	0/1
COI	XS	145	20/57	2/57	0/0
COI	XSD	145	43/57	3/57	0/0
COI	S	193	10/55	0/55	0/0
COI	SD	193	30/55	0/55	0/0
COI	M	218	9/55	0/55	0/0
COI	MD	218	25/55	3/55	0/0
COI	L	293	30/283	0/283	0/0
COI	LD	293	103/283	11/283	0/2
COI	XL	415	11/62	0/62	0/0
COI	XLD	415	37/62	1/62	0/2

Table 7. Released primer sequences

All sequences are written 5' to 3'. Additional-marker endpoints and spans are given in the released per-gene pairs.csv files.

Deg F/R gives the number of concrete variants per direction. The 16S ordinary and D sequences share the reference endpoints in Table 1 and are independently selected at those positions. The single 16S XS pair is retained.

Assay	F, 5'–3'	R, 5'–3'	Deg F/R
XS	AACTTTGGTTGGGGYAAC	TTTCCCCAATCCAACATCGA	2/1
S	CCAGAAAGACGAGAAGACCCTGTC	GAGGAGATAACGCTGTTATCCCTG	1/1
SD	TAAGAAAGACGARAAGACCCTRTC	GAGAAGATGACGCTGTTATCCCTR	4/2
M	AAACTTCTATTAACGTGAAGAA	GTTATCCCTGCGGTAACT	1/1
MD	AAACTTCTATTHAYGTGAAGAA	GTTATCCCTGCGGTAACT	6/1
L	AGACTCGTATGAACGGCAAAACCT	CGCTGTTATCCCTGCGGT	1/1
LD	AGACTTGTATGAAYGGCRAAACMT	CGCTGTTATCCCTGCRGT	8/2
XL	GGTCACGCCTGCCCAGTGGA	AAGATAACGCTGTTATCCCTGCGGTAACTT	1/1
XLD	GGTAACGCCTGCCCARTGGA	GAGATTACGCTGTTATCCCTGCGGTAACTD	2/3

Additional mitochondrial primer sequences

Assay	F, 5'–3'	R, 5'–3'	Deg F/R
12S XS	AGATACCCTACTATACCCTTC	CGACGATGGTATACAAACTG	1/1
12S XSD	AGATACCCTACTATACYYTTC	CGACGATGGTATACWRRCTG	4/8
12S S	AACACCAGAGCAGTACAGCT	CTGCACCTTGATCTGACGT	1/1
12S SD	AACACCAGAGYAGTACARCY	CTGCACCTTGATCTGRHGT	8/6
12S M	AGGAGCTTGTCATTAAATTGAT	TTACTATCGAATCCTCTTTCAT	1/1
12S MD	AGGAGCTTGTCATYRAATYGAT	TTACTATCGAATCCTCTTTCWD	8/6
12S L	AACACCAGAGCAGTACGGCT	GCGATGTGTGCGCATTCTAGAG	1/1
12S LD	AACACCAGAGYAGTACRGYT	GCGATGTGTGCRYATTYTAGAG	8/8
12S XL	GGTGGTTAATAACGTGCCAG	CGACGATGGTATACAGGCTG	1/1
12S XLD	GGTGGTTAATAACGTGCCAG	CGACGATGGTATACWRRCTG	1/8
COI XS	ACTATAGAGGAAAGCAAGAA	GCTGCTGTGAAGTAGGC	1/1
COI XSD	ACTATAGAGGRAAGCMAGAR	GCTGCTGTRAAGTADGC	8/6
COI S	GAATGGTATACGCTATGGT	GTTTCTCAGACTAGCTTTG	1/1
COI SD	GAATGGTATAYGCWATGRT	GTTTCTCAGAYTARCTTKG	8/8
COI M	CACTATAGAGGAAAGCAAGAACC	AACTTTGATCCTTGTAGGGT	1/1
COI MD	CACTATAGAGGWAAGMAAGARCC	AACTTTGATCCTTGTARNGT	8/8
COI L	AAAATGCGAACCCCAGGAGTAAC	GGTTCTTGCTTTCCTCTGTAGTG	1/1
COI LD	AAAATGCGAACYCCAGGARTWAC	GGTTCTTGCTTWCCTCTRTARTG	8/8
COI XL	ATGAACCTAAACCGCTGAT	CGTGGGCTATATTTCTTGA	1/1
COI XLD	ATGAACCTAAAMCGYTGAY	CGTGGGCTATRTTTCTNGA	8/8

Table 8. Paired-primer mismatch sensitivity on the available species panels.

Mismatch ceilings apply to each primer separately. Target counts are species; background counts are accession records under Watson–Crick pairing. The terminal-three column uses the released target compatibility rule. Background ≤2 does not require terminal compatibility and does not replace the broader background models in Tables 2 and 6.

Gene	Assay	Species N	≤0	≤1	≤2	≤2 + terminal 3	Background ≤2
16S	XS	166	0	19	76	75	1
16S	S	174	5	26	110	110	1
16S	SD	174	0	24	130	130	0
16S	M	174	9	34	55	55	0
16S	MD	174	24	51	89	89	0
16S	L	174	13	34	66	65	0
16S	LD	174	29	60	84	82	0
16S	XL	152	1	7	52	23	0
16S	XLD	152	3	15	50	43	0
12S	XS	74	0	4	17	15	0
12S	XSD	74	9	20	33	21	0
12S	S	88	2	5	17	13	0
12S	SD	88	7	16	37	27	0
12S	M	90	0	1	16	8	0
12S	MD	90	4	11	35	28	0
12S	L	83	1	11	20	14	0
12S	LD	83	12	19	36	19	0
12S	XL	53	5	15	31	31	0
12S	XLD	53	7	16	33	33	0
COI	XS	57	2	11	29	20	0
COI	XSD	57	3	18	44	43	0
COI	S	55	0	1	14	10	0
COI	SD	55	0	9	33	30	0
COI	M	55	0	2	13	9	0
COI	MD	55	3	9	25	25	0
COI	L	283	0	12	55	30	0
COI	LD	283	11	59	122	103	0
COI	XL	62	0	1	16	11	0
COI	XLD	62	1	6	37	37	0

Table 9. Primer-free barcode identity on the available window panels.

Unique denotes a species whose ungapped insert differs from every other retained species in the same window. Ambiguous internal inserts are excluded independently of binding-site evaluability. Identical pairs count distinct species pairs within each window.

Gene	Window	Unambiguous N	Unique species	Identical pairs	Insert bp
16S	XS	166	154	6	148–165
16S	S	174	168	3	148–191
16S	M	174	168	3	175–218
16S	L	174	168	3	230–274
16S	XL	152	146	3	313–361
12S	XS	74	70	2	125–147
12S	S	88	86	1	159–172
12S	M	90	88	1	177–190
12S	L	83	81	1	294–310
12S	XL	53	53	0	287–323
COI	XS	57	55	1	104–184
COI	S	55	53	1	69–155
COI	M	55	53	1	102–233
COI	L	278	272	3	247–311
COI	XL	62	62	0	377–377